



BK4819 Technical Reference Manual

TRM-BK4819-E01 V1.0

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1. About This Manual

The BK4819 is a high performance walkie-talkie transceiver. The BK4819 reference manual targets application developers. It provides detailed information on how to program the BK4819.

For pin descriptions, electrical characteristics, package, and ordering information, refer to the corresponding datasheet.

2. MCU Interface - Three-Wire SPI

The BK4819 has a three-wire SPI interface. These three wires are SCK (PIN25), SCN (PIN26), SDATA (PIN27) for data exchange. SCK and SCN are input pins, while SDATA is a bi-direction pin.

Figure 2-1 and Figure 2-2 show the write timing diagram and the read timing diagram of the three-wire SPI.

Figure 2-1 Three-Wire Control Interface Write Timing Diagram

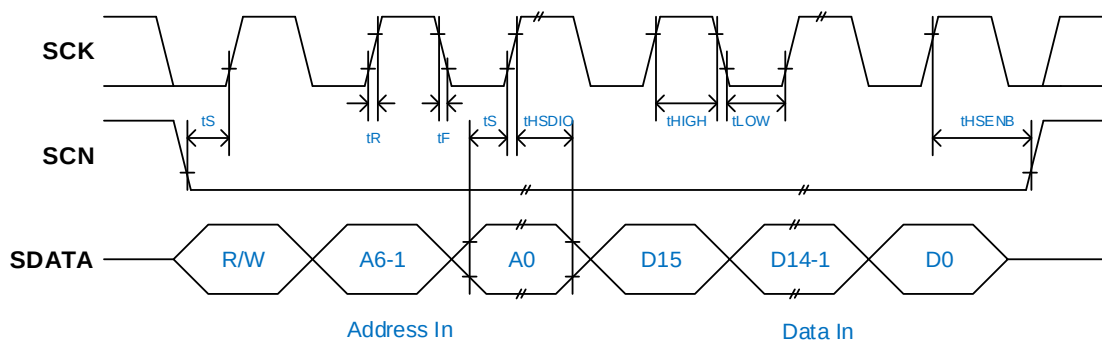


Figure 2-2 Three-Wire Control Interface Read Timing Diagram

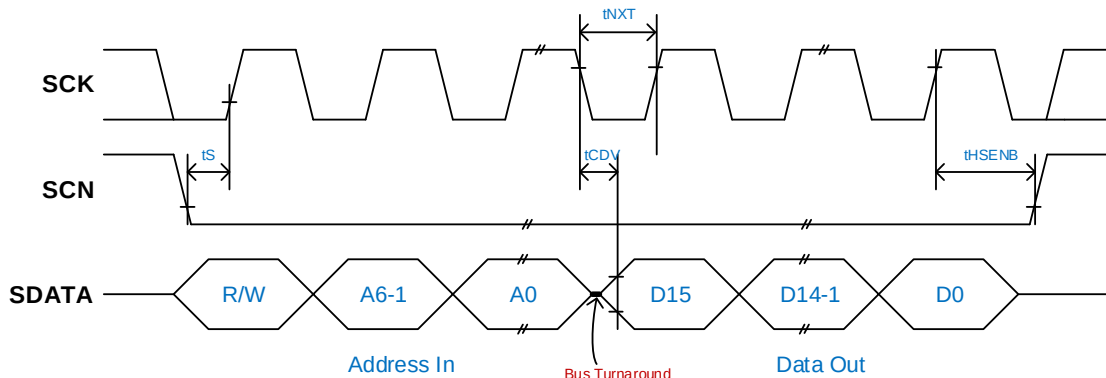


Table 2-1 SPI Control Interface Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
SCK Frequency	f_{SCK}	0	-	8	MHz
SCK High Time	t_{HIGH}	25	-	-	ns
SCK Low Time	t_{LOW}	25	-	-	ns
SDATA Input, SCN to SCK $\uparrow$ Setup	t_S	20	-	-	ns



Parameter	Symbol	Min.	Typ.	Max.	Unit
SDATA Input to SCK↑ Hold	t_{HSDATA}	10	-	-	ns
SCN Input to SCK↓ Hold	t_{HSCN}	10	-	-	ns
SCK↓ to SDATA Output Valid	t_{CDV}	2	-	25	ns
SCK↓ to next SCK↑ after Address In	t_{NXT}	1	-	-	μs
SCK, SCN, SDATA, Rise/Fall Time	t_R, t_F	-	-	10	ns

3. Register Initialization

After power on, the chip performs Soft Reset, internal Power Up and other initialization settings. Call `RF_Initail()` and set parameters including squelch threshold, receive volume, receive AGC, MIC sensitivity, VOX threshold, modulation depth, transmit power, sub-audio, etc. as required for initialization.

4. TX/RX Audio

- After initialization, the normal Speech mode is the default mode, the chip can send and receive voice (300 Hz ~ 3 kHz) in this mode.
- Call RF_EnterCompander() to enable compander and call RF_ExitCompander() to disable compander.

Register	Default	Description
REG_31<3>	0	Enable Compander function: 1: Enable 0: Disable

- Call RF_EnterScramble() to enable scrambling and call RF_ExitScramble() to disable scrambling.

Register	Default	Description
REG_31<1>	0	Enable Scramble function: 1: Enable 0: Disable
REG_71<15:0>	0x8517	Scramble/Tone1 frequency control word: = 3300 (Hz)*10.32444 for XTAL 13M/26M or = 3300 (Hz)*10.48576 for XTAL 12.8M/19.2M/25.6M/38.4M. The scrambler inversion mixing frequency should be kept between 2.6 kHz and 3.5 kHz.

- Enable/disable filters via the registers below:

Register	Default	Description
REG_2B<10>	0	Disable AF RX HPF300 filter: 0: Enable 1: Disable
REG_2B<9>	0	Disable AF RX LPF3K filter: 0: Enable 1: Disable
REG_2B<8>	0	Disable AF RX de-emphasis filter: 0: Enable 1: Disable
REG_2B<2>	0	Disable AF TX HPF300 filter:

Register	Default	Description
		0: Enable 1: Disable
REG_2B<1>	0	Disable AF TX LPF1 filter: 0: Enable 1: Disable
REG_2B<0>	0	Disable AF TX pre-emphasis filter: 0: Enable 1: Disable
REG_43<8:6>	0b001	AF TX LPF2 filter bandwidth (Apass = 1 dB) selection: 100: 4.5 kHz 101: 4.25 kHz 110: 4 kHz 111: 3.75 kHz 000: 3 kHz (for 25k channel space) 001: 2.5 kHz (for 12.5k channel space) 010: 2.75 kHz 011: 3.5 kHz

- Call RF_SetAfResponse(u8 tx, u8 f3k, u8 db) to adjust audio response, where:

Parameter	Value	Description
tx	1	Transmit
	0	Receive
f3k	1	Adjust 3 kHz
	0	Adjust 300 Hz
db	-	Adjustment range: -1 ~ + 4 dB

For example:

- Transmit 300 Hz to increase 2 dB: RF_SetAfResponse(1,0,2);
- Receive 3 kHz to lower 3 dB: RF_SetAfResponse(0,1,-3).

- Set receive volume via the registers below:

Register	Default	Description
REG_48<11:10>	0b00	AF RX Gain1: 00: 0 dB 01: -6 dB 10: -12 dB 11: -18 dB
REG_48<9:4>	0x3C	AF RX Gain2, -26 dB ~ 5.5 dB ,0.5 dB/step: 0x00: Mute
REG_48<3:0>	0b1111	AF DAC Gain (after Gain1 and Gain2), about 2 dB/step: 1111: Max. ... 0000: Min.

- Set transmit modulation and transmit mute via the registers below:

Register	Default	Description
REG_40<12>	1	Enable RF TX deviation: 1: Enable 0: Disable
REG_40<11:0>	0x4D0	RF TX deviation tuning (applicable for both in-band signals and sub-audio signals): 0: Min. ... 0xFFFF: Max.
REG_50<15>	0	Enable AF TX mute (for DTMF TX or other applications): 1: Mute 0: Normal

- Set MIC sensitivity via the register below:

Register	Default	Description
REG_7D<4:0>	0x10	MIC sensitivity tuning, 0.5 dB/step: 0x00: Min. ... 0x1F: Max.



- Call RF_SetAf(u8 mode) to select AF output, call RF_Key() to generate local key tone and prompt tone, and call RF_Call() to generate and transmit ring tone.

5. CTCSS/CDCSS

- Call RF_SetCtcss() and RF_SetCtc2() to enable CTCSS. RF_SetCtc2() is only used to receive CTCSS tails with a frequency of 55 Hz (or other frequencies within 100 Hz); RF_SetCtcss() is used to receive and transmit normal CTCSS.
- Call RF_SetCdcss() and set 134.4 Hz bit rate and CDCSS code to enable CDCSS.
- Call RF_ExitSubau() to disable sub-audio.
- At the end of transmission, call RF_GenTail() to generate tail tone, use parameter CTC120/CTC180/CTC240 for phase tail tone, such as RF_GenTail(CTC180); use parameter CTC55, such as RF_GenTail(CTC55) for channel switch (e.g., 55 Hz); use parameter CTC134 to generate 134.4 Hz tail in CDCSS mode, such as RF_GenTail(CTC134).
- Call RF_GetCtcss() to read CTCSS status, return 1 indicates that CTC1 (main CTC) is received, and return 2 indicates that CTC2 (such as 55 Hz tail tone) is received; call RF_GetCdcss() to read CDCSS status, return 1 indicates that CDC positive code is received, return 2 indicates that CDC negative code is received; call RF_GetTail() to read the phase tail status, return 1 indicates that a 120° phase shift tail tone is received, return 2 indicates that a 180° phase shift tail tone is received, return 3 indicates that a 240° phase shift tail tone is received.

Register	Default	Description
REG_51<15>	0	1: Enable TX CTCSS/CDCSS 0: Disable
REG_51<14>	0	1: GPIO0 Input for CDCSS 0: Normal mode
REG_51<13>	0	1: Transmit negative CDCSS code 0: Transmit positive CDCSS code
REG_51<12>	0	CTCSS/CDCSS mode selection: 1: CTCSS 0: CDCSS
REG_51<11>	0	CDCSS 24 or 23 bit selection: 1: 24 bit 0: 23 bit
REG_51<10>	0	1050 Hz detection mode: 1: 1050/4 detection enable (CTC1 should be set to 1050/4 Hz.)
REG_51<9>	0	Auto CDCSS BW mode: 1: Disable 0: Enable
REG_51<8>	0	Auto CTCSS BW mode:

Register	Default	Description
		0: Enable 1: Disable
REG_51<6:0>	0	CTCSS/CDCSS TX Gain1 tuning: 0: Min. ... 0x7F: Max.
REG_2E<9:8>	0x10	CTCSS/CDCSS TX Gain2 tuning (after Gain1): 00: 12 dB 01: 6 dB 10: 0 dB 11: -6 dB
REG_07<15:0>	-	<15:13>: 0: CTC1 <12:0>: CTC1 frequency control word: = freq (Hz)*20.64888 for XTAL 13M/26M or = freq (Hz)*20.97152 for XTAL 12.8M/19.2M/25.6M/38.4M <15:13>: 1: CTC2 (Tail 55 Hz RX detection) <12:0>: CTC2 (should below 100 Hz) frequency control word: = 25391/freq (Hz) for XTAL 13M/26M or = 25000/freq (Hz) for XTAL 12.8M/19.2M/25.6M/38.4M <15:13>: 2: CDCSS 134.4 Hz <12:0>: CDCSS baud rate frequency (134.4 Hz) control word: = freq (Hz)*20.64888 for XTAL 13M/26M or = freq (Hz)*20.97152 for XTAL 12.8M/19.2M/25.6M/38.4M
REG_08<15:0>	-	<15>: 1: CDCSS high 12 bits; 0: CDCSS low 12 bits <11:0>: CDCSS high/low 12 bit code
REG_52<15>	0	Enable 120°/180°/240° shift CTCSS or 134.4 Hz Tail when CDCSS mode is enabled: 0: Normal 1: Enable
REG_52<14:13>	0b00	CTCSS Tail mode selection (only valid when REG_52<15> = 1): 00: 134.4 Hz CTCSS Tail when CDCSS mode is enabled 01: CTCSS0 120 °phase shift 10: CTCSS0 180 °phase shift

Register	Default	Description
		11: CTCSS0 240 °phase shift
REG_52<12>	0	CTCSS detection threshold mode: 1: ~ 0.1% 0: 0.1 Hz
REG_52<11:6>	0x0A	CTCSS Found detection threshold
REG_52<5:0>	0x0F	CTCSS Lost detection threshold
REG_0C<15:14>	Read Only	<15>: CDCSS negative code received <14>: CDCSS positive code received
REG_0C<13:12>	Read Only	CTCSS phase shift received: 00: No phase shift 01: CTCSS0 120 °phase shift 10: CTCSS0 180 °phase shift 11: CTCSS0 240 °phase shift
REG_0C<11:10 >	Read Only	<11>: CTC2 (55 Hz) received <10>: CTC1 received

6. SELCALL

- Call RF_Enter5tone() to enable SELCALL(5Tone) mode. This function only sets the receive frequency coefficient, receive threshold, and transmit channel, and does not affect normal audio listening.
- Call RF_Exit5tone() to exit SELCALL(5Tone) mode.
- Change the SELCALL frequency (Tone1) in time according to the transmit code and call RF_5toneTransmit() to transmit SELCALL(5Tone).
- Call RF_5toneReceive() to receive SELCALL(5Tone), return 1 means failure and return 0 means success.

7. DTMF

- Call RF_EnterDtmf() to enable DTMF mode. This function only sets the DTMF receive frequency coefficient, receive threshold, and transmit path, and does not affect normal audio listening.
- Call RF_ExitDtmf() to exit DTMF mode.
- Change the DTMF frequency (Tone1+Tone2) in time according to the transmit code and call RF_DtmfTransmit() to transmit DTMF.
- Call RF_DtmfReceive() to receive DTMF, return 1 means failure and return 0 means success.

Register	Default	Description
REG_70<15>	0	Enable TONE1: 1: Enable 0: Disable
REG_70<14:8>	0	TONE1 tuning gain
REG_70<7>	0	Enable TONE2: 1: Enable 0: Disable
REG_70<6:0>	0	TONE2/FSK tuning gain
REG_71<15:0>	0x8517	TONE1/Scramble frequency control word: = freq (Hz)*10.32444 for XTAL 13M/26M or = freq (Hz)*10.48576 for XTAL 12.8M/19.2M/25.6M/38.4M
REG_72<15:0>	0x2854	TONE2/FSK frequency control word: = freq (Hz)*10.32444 for XTAL 13M/26M or = freq (Hz)*10.48576 for XTAL 12.8M/19.2M/25.6M/38.4M
REG_50<15>	0	Enable AF TX mute (for DTMF TX or other applications): 1: Mute 0: Normal
REG_0B<11:8>	Read Only	DTMF/5 Tone code received

8. FSK

- Calling RF_EnterFsk() to enable FSK mode will not affect normal audio listening and the BK4819 can enter DTMF/SELCALL mode for receive at the same time. The FSK rate register is multiplexed with the Tone2 register. To use 2400 bps mode, you need to enable the macro FSK 2.4K.
- Call RF_ExitFsk() to exit FSK mode.
- Call RF_FskTransmit() to transmit FSK, return 1 means failure and return 0 means success.
- FSK frame format (CRC is optional): To ensure compatibility with BK4815/BK4818, you need to complete the Addr/Type/Size/CRCA/Payload/CRCB data framing of the BK4815/BK4818 FSK frame structure in the Data, disable the CRC of the BK4819 FSK frame structure, and set the same Preamble and Sync Word.
- Call RF_FskReceive() to receive FSK, return 1 means failure and return 0 means success.

Preamble	Sync Word	Data	CRC (opt)
1 ~ 16 bytes	2 or 4 bytes	Config (maximum 1024 words) 1 word = 2 bytes	2 bytes

Register	Default	Description
REG_58<15:13>	000	FSK TX mode selection: 000: FSK 1.2K and FSK 2.4K TX 001: FFSK 1.2K/1.8K TX 011: FFSK 1.2K/2.4K TX 101: NOAA SAME TX
REG_58<12:10>	000	FSK RX mode selection: 000: FSK 1.2K, FSK 2.4K RX and NOAA SAME RX 111: FFSK 1.2K/1.8K RX 100: FFSK 1.2K/2.4K RX
REG_58<9:8>	00	FSK RX Gain
REG_58<7:6>	00	11: FSK enable 00: FSK disable
REG_58<5:4>	00	FSK Preamble type selection: 11: 0xAA 10: 0x55 00: 0xAA or 0x55 due to the MSB of FSK Sync Byte 0

Register	Default	Description
REG_58<3:1>	000	FSK RX bandwidth setting: 100: FSK 2.4K and FFSK 1.2K/2.4K 000: FSK 1.2K 001: FFSK 1.2K/1.8K 010: NOAA SAME RX
REG_58<0>	0	FSK enable: 1: Enable 0: Disable
REG_59<15>	0	Clear TX FIFO: 1: Clear
REG_59<14>	0	Clear RX FIFO: 1: Clear
REG_59<13>	0	1: Enable FSK Scramble
REG_59<12>	0	1: Enable FSK RX
REG_59<11>	0	1: Enable FSK TX
REG_59<10>	0	1: Invert FSK data when RX mode is enabled
REG_59<9>	0	1: Invert FSK data when TX mode is enabled
REG_59<7:4>	0	FSK Preamble length selection: 0: 1 byte 1: 2 bytes 2: 3 bytes ... 15: 16 bytes
REG_59<3>	0	FSK Sync length selection: 1: 4 bytes (FSK Sync Byte 0, 1, 2, 3) 0: 2 bytes (FSK Sync Byte 0, 1)
REG_5A<15:8>	0x85	FSK Sync Byte 0 (Sync Byte 0 first, then Byte 1, 2, 3)
REG_5A<7:0>	0xCF	FSK Sync Byte 1
REG_5B<15:8>	0xAB	FSK Sync Byte 2
REG_5B<7:0>	0x45	FSK Sync Byte 3
REG_5C<6>	1	CRC option enable:

Register	Default	Description
		1: Enable 0: Disable
REG_5D<15:8>	0x0F	FSK data length (byte) low 8 bits. For example, 0xF means 16-byte length.
REG_5D<7:5>	0	FSK data length (byte) high 3 bits
REG_5E<9:3>	64	FSK TX FIFO (total 128 words) Almost Empty threshold
REG_5E<2:0>	4	FSK RX FIFO (total 8 words) Almost Full threshold
REG_5F<15:0>	x	FSK word input/output
REG_70<6:0>	0	TONE2/FSK tuning gain
REG_72<15:0>	0x2854	TONE2/FSK frequency control word: = freq (Hz)*10.32444 for XTAL 13M/26M or = freq (Hz)*10.48576 for XTAL 12.8M/19.2M/25.6M/38.4M
REG_0B<7>	Read Only	FSK RX SyncN Found
REG_0B<6>	Read Only	FSK RX SyncP Found
REG_0B<4>	Read Only	FSK RX CRC indicator: 1: CRC pass 0: CRC fail

9. MDC1200

- Calling RF_EnterMdc() to enable MDC mode will not affect normal audio listening and the BK4819 can enter DTMF/SELCALL mode for receive at the same time. The MDC rate register is multiplexed with the Tone2 register. The default mode is 1.2K/1.8K mode, if you want to use 1.2K/2.4K mode, you need to enable the macro MDC2400.
- Call RF_ExitMdc() to exit MDC mode.
- Call RF_MdcTransmit() to transmit MDC, return 1 means failure and return 0 means success.
- Call RF_MdcReceive() to receive MDC, return 1 means failure and return 0 means success.
- Supports HDC1200 mode.

10. NOAA SAME

- Calling RF_EnterNoaa() to receive NOAA SAME code will not affect normal audio listening and the BK4819 can enter NOAA mode for receive at the same time. The NOAA rate register is multiplexed with the Tone2 register.
- Call RF_ExitNoaa() to disable NOAA SAME code receive.
- Call RF_NoaaReceive() to receive FSK, return 1 means failure and return 0 means success. The MCU performs post processing according to the received code.

11. VOX

- Call RF_EnterVox() to enable VOX; call RF_ExitVox() to disable VOX.
- Call RF_GetVox() to get VOX status, return 1 means MIC voice received and return 0 means no voice received.
- Call RF_GetVoxAmp() to get the VOX amplitude, and the return value is the MIC voice amplitude, which is used by the MCU to determine whether to activate the VOX.

Register	Default	Description
REG_31<2>	0	Enable VOX detection: 1: Enable 0: Disable
REG_7A<15:12>	8	VOX = 0 detection delay, ~ 128 ms
REG_46<10:0>	0x50	Voice amplitude threshold for VOX on
REG_79<15:11>	8	VOX detection interval time
REG_79<10:0>	0x40	Voice amplitude threshold for VOX off
REG_64<15:0>	Read Only	Voice amplitude out
REG_0C<2>	Read Only	VOX indicator: 0: No 1: Yes

12. Power Saving

- Call RF_Sleep() to enter sleep, call RF_WakeUp() for wakeup, call RF_Txon() to enter transmit mode and call RF_Rxon() to enter receive mode.
- The SLEEP current is about 200 μ A ~ 300 μ A; after wakeup, the IDLE current is about 3 mA.

Register	Default	Description
REG_37<14:12>	0b001	DSP voltage setting
REG_37<11>	1	ANA LDO selection: 1: 2.7 V 0: 2.4 V
REG_37<10>	1	VCO LDO selection: 1: 2.7 V 0: 2.4 V
REG_37<9>	1	RF LDO selection: 1: 2.7 V 0: 2.4 V
REG_37<8>	1	PLL LDO selection: 1: 2.7 V 0: 2.4 V
REG_37<7>	0	ANA LDO Bypass: 1: Bypass 0: Enable
REG_37<6>	0	VCO LDO Bypass: 1: Bypass 0: Enable
REG_37<5>	0	RF LDO Bypass: 1: Bypass 0: Enable
REG_37<4>	0	PLL LDO Bypass: 1: Bypass 0: Enable
REG_37<3>	0	Reserved
REG_37<2>	0	DSP enable:

Register	Default	Description
		1: Enable 0: Disable
REG_37<1>	0	XTAL enable: 1: Enable 0: Disable
REG_37<0>	0	Band Gap enable: 1: Enable 0: Disable

13. TX/RX Mode Switch

- Call RF_Txon() to enter transmit mode.
- Call RF_Rxon() to enter receive mode.
- Call RF_Txon_Beep() to transmit the ring tone.

Register	Default	Description
REG_30<15>	0	VCO calibration enable: 1: Enable 0: Disable
REG_30<13:10>	0	RX link enable (includes LNA/MIXER/PGA/ADC): 1111: Enable 0000: Disable
REG_30<9>	0	AF DAC enable: 1: Enable 0: Disable
REG_30<7:4>	0	PLL/VCO enable: 1111: Enable 0000: Disable
REG_30<3>	0	PA Gain enable: 1: Enable 0: Disable
REG_30<2>	0	MIC ADC enable: 1: Enable 0: Disable
REG_30<1>	0	TX DSP enable: 1: Enable 0: Disable
REG_30<0>	0	RX DSP enable: 1: Enable 0: Disable

14. Squelch, RSSI, Ex-Noise, Glitch

You can get corresponding parameters through RF_GetRssi(), RF_GetNoise(), RF_GetGlitch() for setting the squelch level.

Register	Default	Description
REG_78<15:8>	0x48	RSSI threshold for Squelch = 1, 0.5 dB/step
REG_78<7:0>	0x46	RSSI threshold for Squelch = 0, 0.5 dB/step
REG_4F<14:8>	0x2F	Ex-noise threshold for Squelch = 0
REG_4F<6:0>	0x2E	Ex-noise threshold for Squelch = 1
REG_4D<7:0>	0x20	Glitch threshold for Squelch = 0
REG_4E<7:0>	0x08	Glitch threshold for Squelch = 1
REG_4E<13:11>	0b101	Squelch = 1 Delay Setting (default: 5 ms)
REG_4E<10:9>	0b111	Squelch = 0 Delay Setting (default: 5 ms)
REG_67<8:0>	Read Only	RSSI (dBm) = REG_67<8:0>/2 – 160, 0.5 dB/step
REG_65<6:0>	Read Only	Ex-noise indicator, dB/step
REG_63<7:0>	Read Only	Glitch indicator
REG_0C<1>	Read Only	Squelch result output: 1: Link 0: Loss

15. AFC, ALC, MIC AGC

The MIC PGA gain automatic control can expand the dynamic range of the MIC signal, so that the MIC signal of larger amplitude can be transmitted without distortion. This function should be disabled in the DMR solution (REG_19<15> = 1), and the ALC function should also be disabled (REG_4B<5> = 1).

Register	Default	Description
REG_73<13:11>	0b000	Automatic Frequency Correction (AFC) range selection: 000: Max. ... 111: Min.
REG_73<4>	0	Automatic Frequency Correction (AFC) disable: 1: Disable 0: Enable
REG_19<15>	1	Automatic MIC PGA Gain Controller (MIC AGC) disable: 1: Disable 0: Enable
REG_4B<5>	0	AF Level Controller (ALC) disable: 1: Disable 0: Enable

16. Frequency Setting

Call RF_SetFreq(u16 freq_hi16, u16 freq_lo16) to set the frequency, the conversion formula is as below: Frequency (Hz) = (freq_hi16<<16 + freq_lo16)*10

Register	Default	Description
REG_38<15:0>	0x3A98	Frequency (Hz) = (freq_hi16<<16 + freq_lo16)*10
REG_39<15:0>	0x0271	

For example, to set a channel frequency of 409.75 MHz, call RF_SetFreq((40975000>>16)&0xFFFF, 40975000&0xFFFF)

17. TX Output Power

Register	Default	Description
REG_36<15:8>	0	PA bias output 0 ~ 3.2 V: 0x00: 0 V ... 0xFF: 3.2 V
REG_36<7>	0	1: Enable PA CTL output 0: Disable (0 V output)
REG_36<5:3>	0b111	PA Gain1 tuning: 111: Max. ... 000: Min.
REG_36<2:0>	0b111	PA Gain2 tuning: 111: Max. ... 000: Min.

Table 17-1 Output Power Table

Power (dBm)								
PA Gain2	PA Gain1							
	111	110	101	100	011	010	001	000
111	7.26	7.01	6.67	6.17	5.39	4.10	1.52	-5.02
110	6.38	6.08	5.67	6.06	4.12	2.53	-0.72	-9.45
101	5.65	5.30	4.82	4.13	3.03	1.18	-2.58	-13.4
100	5.01	4.62	4.08	3.30	2.08	0.01	-4.16	-16.9
011	4.19	3.73	3.11	2.21	0.84	-1.48	-6.11	-20.5
010	2.60	2.04	1.24	0.13	-1.56	-4.25	-9.50	-22.9
001	1.04	0.35	-0.63	-1.98	-3.90	-6.80	-12.3	-23.7
000	-0.70	-1.51	-2.65	-4.16	-6.20	-9.35	-14.9	-24.3

18. Interrupt

- The interrupt signal can be sent by any GPIO (see the GPIO setting function in reference code), or query the REG_0C<0>, active high, low by default.
- Interrupt can be output by any GPIO port of the chip. The interrupt can be cleared by writing any value to the register REG_02, such as: RF_Write (0x02,0x0000); //clear interrupt.
- The interrupt is active high (Or the interrupt register REG_0C<0> is 1). When the interrupt is triggered, the interrupt must be cleared before the interrupt vector table can be read.

Register	Default	Description
REG_0C<0>	Read Only	Interrupt indicator: 1: Interrupt request 0: No request
REG_3F<15>	0	FSK TX Finished interrupt enable: 1: Enable 0: Disable
REG_3F<14>	0	FSK FIFO Almost Empty interrupt enable: 1: Enable 0: Disable
REG_3F<13>	0	FSK RX Finished interrupt enable: 1: Enable 0: Disable
REG_3F<12>	0	FSK FIFO Almost Full interrupt enable: 1: Enable 0: Disable
REG_3F<11>	0	DTMF/5 TONE Found interrupt enable: 1: Enable 0: Disable
REG_3F<10>	0	CTCSS/CDCSS Tail Found interrupt enable: 1: Enable 0: Disable
REG_3F<9>	0	CDCSS Found interrupt enable: 1: Enable 0: Disable

Register	Default	Description
REG_3F<8>	0	CDCSS Lost interrupt enable: 1: Enable 0: Disable
REG_3F<7>	0	CTCSS Found interrupt enable: 1: Enable 0: Disable
REG_3F<6>	0	CTCSS Lost interrupt enable: 1: Enable 0: Disable
REG_3F<5>	0	VOX Found interrupt enable: 1: Enable 0: Disable
REG_3F<4>	0	VOX Lost interrupt enable: 1: Enable 0: Disable
REG_3F<3>	0	Squelch Found interrupt enable: 1: Enable 0: Disable
REG_3F<2>	0	Squelch Lost interrupt enable: 1: Enable 0: Disable
REG_3F<1>	0	FSK RX Sync interrupt enable: 1: Enable 0: Disable
REG_02<15>	Read Only	FSK TX Finished interrupt
REG_02<14>	Read Only	FSK FIFO Almost Empty interrupt
REG_02<13>	Read Only	FSK RX Finished interrupt
REG_02<12>	Read Only	FSK FIFO Almost Full interrupt
REG_02<11>	Read Only	DTMF/5 TONE Found interrupt
REG_02<10>	Read Only	CTCSS/CDCSS Tail Found interrupt
REG_02<9>	Read Only	CDCSS Found interrupt



Register	Default	Description
REG_02<8>	Read Only	CDCSS Lost interrupt
REG_02<7>	Read Only	CTCSS Found interrupt
REG_02<6>	Read Only	CTCSS Lost interrupt
REG_02<5>	Read Only	VOX Found interrupt
REG_02<4>	Read Only	VOX Lost interrupt
REG_02<3>	Read Only	Squelch Found interrupt
REG_02<2>	Read Only	Squelch Lost interrupt
REG_02<1>	Read Only	FSK RX Sync interrupt

19. GPIO

- Call RF_SetGpioOut(u8 num, u8 type, u8 val) according to corresponding PINs and output mode, where num is the internal number, type is the output mode, and val is the output value in the GPIO output mode. The output mode is detailed in the reference code driver.c.
- Call RF_GetGpioIn(u8 num) to get the GPIO input value, where num is the internal number.

Register	Default	Description
REG_0A<6>	Read Only	GPIO6 input indicator: 1: High 0: Low
REG_0A<5>	Read Only	GPIO5 input indicator: 1: High 0: Low
REG_0A<4>	Read Only	GPIO4 input indicator: 1: High 0: Low
REG_0A<3>	Read Only	GPIO3 input indicator: 1: High 0: Low
REG_0A<2>	Read Only	GPIO2 input indicator: 1: High 0: Low
REG_0A<1>	Read Only	GPIO1 Input Indicator: 1: High 0: Low
REG_0A<0>	Read Only	GPIO0 input indicator: 1: High 0: Low
REG_33<14>	1	GPIO6 output disable: 1: Output disable 0: Output enable
REG_33<13>	1	GPIO5 output disable: 1: Output disable

Register	Default	Description
		0: Output enable
REG_33<12>	1	GPIO4 output disable: 1: Output disable 0: Output enable
REG_33<11>	1	GPIO3 output disable: 1: Output disable 0: Output enable
REG_33<10>	1	GPIO2 output disable: 1: Output disable 0: Output enable
REG_33<9>	1	GPIO1 output disable: 1: Output disable 0: Output enable
REG_33<8>	1	GPIO0 output disable: 1: Output disable 0: Output enable
REG_33<6>	0	GPIO6 output value: 1: High when output is enabled 0: Low when output is enabled
REG_33<5>	0	GPIO5 output value: 1: High when output is enabled 0: Low when output is enabled
REG_33<4>	0	GPIO4 output value: 1: High when output is enabled 0: Low when output is enabled
REG_33<3>	0	GPIO3 output value: 1: High when output is enabled 0: Low when output is enabled
REG_33<2>	0	GPIO2 output value: 1: High when output is enabled 0: Low when output is enabled
REG_33<1>	0	GPIO1 output value: 1: High when output is enabled

Register	Default	Description
		0: Low when output is enabled
REG_33<0>	0	GPIO0 output value: 1: High when output is enabled 0: Low when output is enabled
REG_34<15:12>	0x0	GPIO3 output type selection: 0: High/Low 1: Interrupt 2: Squelch 3 = VOX 4: CTCSS/CDCSS comparison results 5: CTCSS comparison results 6: CDCSS comparison results 7: Tail detection results 8: DTMF/5 Tone symbol received flag 9: CTCSS/CDCSS digital wave Others: Reserved
REG_34<11:8>	0x0	GPIO2 output type selection. The description is the same as REG_34<15:12>.
REG_34<7:4>	0x0	GPIO1 output type selection. The description is the same as REG_34<15:12>.
REG_34<3:0>	0x0	GPIO0 output type selection The description is the same as REG_34<15:12>.
REG_35<11:8>	0x0	GPIO6 output type selection. The description is the same as REG_34<15:12>.
REG_35<7:4>	0x0	GPIO5 output type selection. The description is the same as REG_34<15:12>.
REG_35<3:0>	0x0	GPIO4 output type selection. The description is the same as REG_34<15:12>.

20. XTAL

The chip supports crystals or temperature compensated crystals of 26 MHz, 25.6 MHz, 13 MHz, 12.8 MHz, 19.2 MHz and 38.4 MHz. The default crystal is 26 MHz. If you use a crystal or a temperature compensated crystal of frequencies other than 26 MHz, call `RF_SetXtal(u8 mode)`, for example, `RF_SetXtal(XTAL19M2)`.

21. Frequency Scan

- Call RF_FreqScan() to scan the frequency to obtain the RF frequency of the LNAIN pin (requires a larger amplitude > -40 dBm). Return 1 means failure and return 0 means success. The frequency is written to the global variables FRQ_HI16 and FRQ_LO16.
- After scanning the frequency, set the receiving frequency to this frequency, and call RF_CtcDcsScan() to obtain the CTCSS frequency or CDCSS code. Return 0 means failure, return 1 means that CTCSS is received and the frequency is written to the global variable CtC_FREQ, and return 2 means that CDCSS is received and the frequency is written to the global variables DCS_HI12 and DCS_LO12.

Register	Default	Description
REG_32<15:14>	0b00	Frequency scan time: 00: 0.2 s 01: 0.4 s 10: 0.8 s 11: 1.6 s
REG_32<0>	0	Frequency scan enable: 1: Enable 0: Disable
REG_0D<15>	Read Only	Frequency scan indicator: 1: Busy 0: Finished
REG_0D<10:0>	Read Only	Frequency scan high 16 bits
REG_0E<15:0>	Read Only	Frequency scan low 16 bits: = REG_0D<10:0><<16 + REG_0E<15:0> (unit: 10 Hz)
REG_68<15>	Read Only	CTCSS scan indicator: 1: Busy 0: Found
REG_68<12:0>	Read Only	CTCSS frequency. Frequency (Hz): = REG_68<12:0>/20.64888 for 13M/26M XTAL or = REG_68<12:0>/20.97152 for 12.8M/19.2M/25.6M/38.4M XTAL
REG_69<15>	Read Only	CDCSS scan indicator: 1: Busy 0: Found

Register	Default	Description
REG_69<14>	Read Only	23 or 24 bit CDCSS indicator: 1: 24 bit 0: 23 bit
REG_69<11:0>	Read Only	CDCSS high 12 bits
REG_6A<11:0>	Read Only	CDCSS low 12 bits

22. Channel Spacing

The chip supports a variety of bandwidths, including common 12.5 kHz, 25 kHz, 6.25 kHz and 20 kHz. Call RF_SetChnSpace (u8 space) and input parameters SPACE_12K5/SPACE_25K/SPACE_6K25/SPACE_20K according to actual needs to set the transmit and receive bandwidth.

Register	Default	Description
REG_43<14:12>	0b100	RF filter bandwidth (Apass = 0.1 dB): 000: 1.7 kHz 001: 2 kHz 010: 2.5 kHz 011: 3 kHz 100: 3.75 kHz 101: 4 kHz 110: 4.25 kHz 111: 4.5 kHz If REG_43<5> = 1, RF filter BW = BW*2.
REG_43<11:9>	0b000	RF filter bandwidth when the signal is weak (Apass = 0.1 dB): 000: 1.7 kHz 001: 2 kHz 010: 2.5 kHz 011: 3 kHz 100: 3.75 kHz 101: 4 kHz 110: 4.25 kHz 111: 4.5 kHz If REG_43<5> = 1, RF filter BW = BW*2.
REG_43<8:6>	0b001	AF TX LPF2 filter bandwidth (Apass = 1 dB) selection: 100: 4.5 kHz 101: 4.25 kHz 110: 4 kHz 111: 3.75 kHz 000: 3 kHz (for 25k channel space) 001: 2.5 kHz (for 12.5k channel space) 010: 2.75 kHz 011: 3.5 kHz



Register	Default	Description
REG_43<5:4>	0b00	Bandwidth mode selection: 00: 12.5k 01: 6.25k 10: 25k/20k

23. Digital Walkie-Talkie

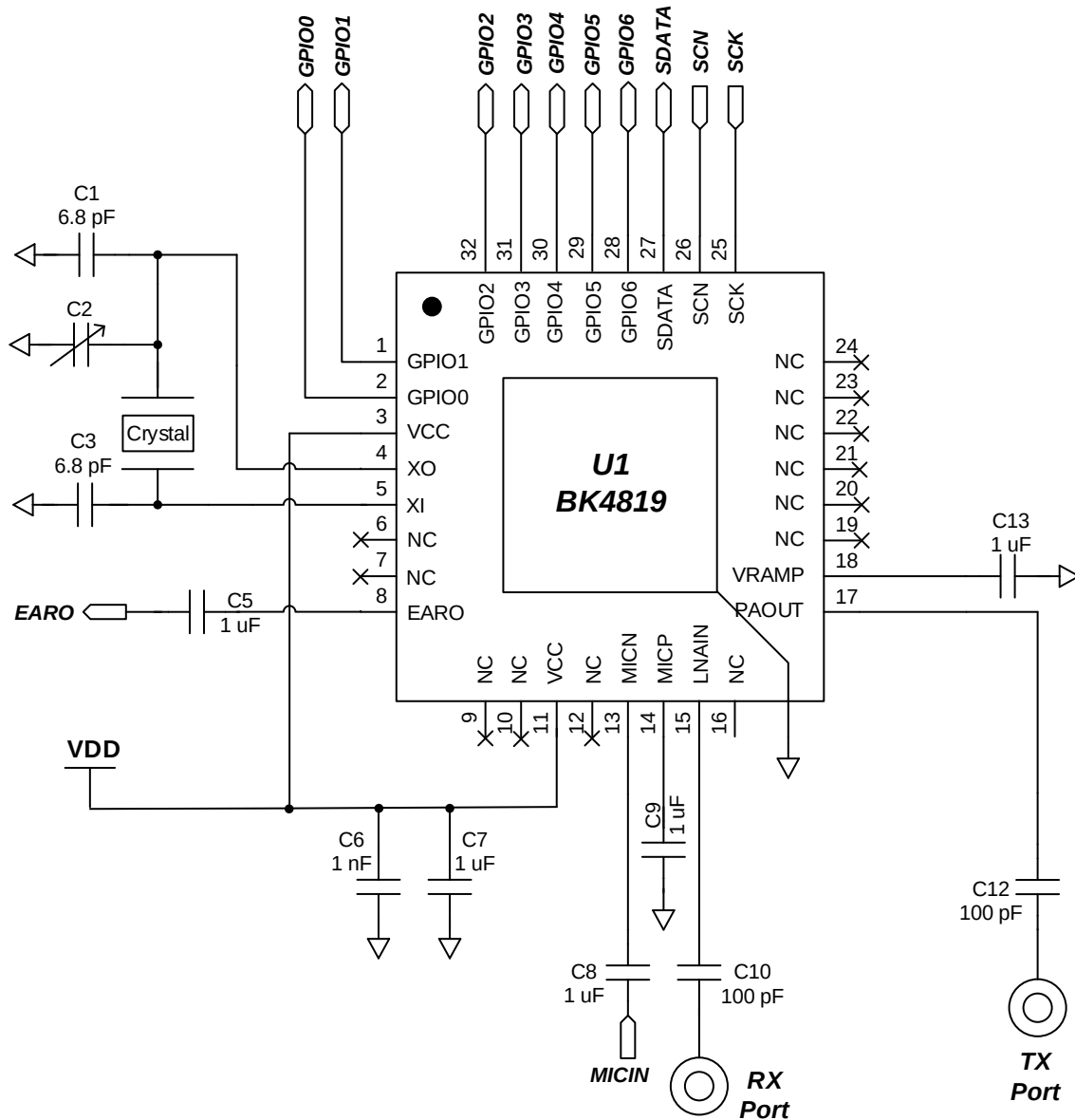
When the chip is used in a digital walkie-talkie, all audio filters need to be bypassed. Call `RF_EnterBypass()` to bypass audio filters and call `RF_ExitBypass()` to exit this mode. The reference code is as below:

```
void RF_EnterBypass() //for dMR
{
    RF_Write(0x47, REG_47 | 9<<8 | 3); //[11:8]=9 for Rx; [1]=1 for gain; [0]=1 for Tx
    RF_Write(0x7E, REG_7E & 0xFFC7); //[5:3]=000b for Tx DCC bypass
}

void RF_ExitBypass()
{
    RF_SetAf(MUTE);
    RF_Write(0x7E, REG_7E);
}
```

24. Hardware Design

Figure 24-1 BK4819 Application Schematic





Revision History

Version	Date	Description
1.0	2022/3/21	Initial release.

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Beken Corporation

Building 41, 1387 Zhangdong Rd
Shanghai 201203
China

<http://www.bekencorp.com>